Chahat Raj

PhD Candidate in Computer Science | George Mason University

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Research Interests

My research focuses on NLP, particularly the evaluation and alignment of large language models. I study socio-cultural biases and other harms in multilingual and multimodal systems and develop methods to mitigate them, advancing trustworthy and responsible AI. My work has been published at ACL, EMNLP, AIES, AAAI, and other venues.

Education

May 2027 | **George Mason University** — **Fairfax, VA**
Aug 2022 | Ph.D. in Computer Science
Advisors: **Ziwei Zhu**, **Antonios Anastasopoulos**

Dec 2025 | **George Mason University** — **Fairfax, VA**
Aug 2022 | Masters in Computer Science

Aug 2021 | **Delhi Technological University** — **Delhi, India**
Aug 2019 | Masters in Information Systems (Thesis)
Advisor: **Priyanka Meel**

May 2019 | **Indira Gandhi Delhi Technical University for Women** — **Delhi, India**
Aug 2015 | Bachelors in Computer Science & Engineering

Experience

Aug 2026 | **Amazon** — **Seattle, WA**
May 2026 | Applied Scientist Intern | Mentor: **Ren Pang**, Manager: **Zheng Qi**
› *A parameter-efficient safety steering technique for bedrock guardrails foundation models.*

Aug 2025 | **George Mason University, University of Washington** — **Fairfax, VA**
May 2024 | Graduate Research Assistant | Advisors: **Ziwei Zhu**, **Antonios Anastasopoulos**, **Aylin Caliskan**
› *Societal simulations, model unlearning, and cognitive alignment for bias mitigation.*

Sep 2024 | **University of Washington | Tech Policy Lab, The Information School** — **Seattle, WA**
May 2024 | Visiting Scholar | Advisor: **Aylin Caliskan**
› *Bias, cognition, and mitigation in vision-language models.*

Aug 2023 | **George Mason University | School of Computer Science** — **Fairfax, VA**
May 2023 | Graduate Research Assistant | Advisor: **Ziwei Zhu**
› *Fairness, interpretability, and robustness in transformer-based fake news detection.*

Jul 2022 | **University of Technology Sydney | School of Computer Science** — **Sydney, AU**
Jun 2021 | Visiting Scholar | Advisor: **Mukesh Prasad**
› *Cyberbullying, AI for Aboriginals, modeling disaster fundraising, and ML for clinical diagnosis.*

Selected Publications

full list here

[C.12] **VIGNETTE: Socially Grounded Bias Evaluation for Vision-Language Models** [PDF | Code] **SAC Highlight Award**
Chahat Raj, Bowen Wei, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Proceedings of the Association for Computational Linguistics: ACL '26, San Diego, California, USA [**ACL Main '26**]

[C.11] **Talent or Luck? Evaluating Attribution Bias in Large Language Models** [PDF | Code]
Chahat Raj, Mahika Banerjee, Jinhao Pan, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Findings of the Association for Computational Linguistics: ACL '26, San Diego, California, USA [**ACL Findings '26**]

[C.10] **Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement** [PDF | Code]
Jinhao, **Chahat Raj**, Anjishnu Mukherjee, Sina Mansouri, Bowen Wei, Shloka Yada, Ziwei Zhu
International Conference in Machine Learning: ICML '26, Seoul, South Korea [**ICML '26**]

[C.9] **Measuring South Asian Biases in Large Language Models** [PDF | Code]
Mamnuya Rinki, **Chahat Raj**, Anjishnu Mukherjee, Ziwei Zhu
TrustNLP at ACL '26 [**TrustNLP '26**]

[C.8] **Discovering Bias Associations through Open-Ended LLM Generations** [PDF | Code]
Jinhao Pan, **Chahat Raj**, Ziwei Zhu
In Proceedings of the AAAI Conference on Artificial Intelligence [**AAAI '26**]

[C.7] **What's Not Said Still Hurts: A Description-Based Evaluation Framework for Measuring Social Bias in LLMs**
Jinhao Pan, **Chahat Raj**, Ziyu Yao, Ziwei Zhu [PDF | Code]
In Findings of the Association for Computational Linguistics: EMNLP '25, Suzhou, China [**EMNLP Findings '25**]

[C.6] **Toward Inclusive Language Models: Sparsity-Driven Calibration for Systematic and Interpretable Mitigation of Social Biases in LLMs** [PDF | Code]
Prommy Sultana Hossain, **Chahat Raj**, Ziwei Zhu, Jessica Lin, Emanuela Marasco
In Findings of the Association for Computational Linguistics: EMNLP '25, Suzhou, China [**EMNLP Findings '25**]

[C.5] **BiasDora: Exploring Hidden Biased Associations in Vision-Language Models** [PDF | Code]
Chahat Raj, Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Findings of the Association for Computational Linguistics: EMNLP '24, Miami, Florida, USA [**EMNLP Findings '24**]

[C.4] **Breaking Bias, Building Bridges: Evaluation and Mitigation of Social Biases in LLMs via Contact Hypothesis**
Chahat Raj, Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu [PDF | Code]
Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society [**AIES '24**]

[C.3] **Global Voices, Local Biases: Socio-cultural Prejudices across Languages** [PDF | Code]
Chahat Raj*, Anjishnu Mukherjee*, Ziwei Zhu, Antonios Anastasopoulos
In Proceedings of EMNLP '23, Singapore [**EMNLP Main '23**]

[C.2] **SALSA: Salience-Based Switching Attack for Adversarial Perturbations in Fake News Detection Models**
Chahat Raj*, Anjishnu Mukherjee*, Hemant Purohit, Antonios Anastasopoulos, Ziwei Zhu [PDF | Code]
European Conference on Information Retrieval [**ECIR '24**]

[C.1] **True and Fair: Robust and Unbiased Fake News Detection via Interpretable Machine Learning** [PDF | Code]
Chahat Raj, Anjishnu Mukherjee, Ziwei Zhu
Proceedings of the AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society [**AIES '23**]

Under Review

[S.2] **Debias It Yourself: Teaching LLMs Cognitive Bias-Mitigation Interventions** [PDF | Code]
Chahat Raj, Sina Mansouri, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
[**In Submission**]

[S.1] **Inoculation Prompting: Implicit Bias Mitigation in Large Language Models**
Yongxu Sun, **Chahat Raj**, Aylin Caliskan
[**In Submission**]

Selected Projects

› **AI, Power & Society**	Auditing Diffusion Models for Generational and Representational Errors	[Report]
› **Computer Vision**	Cultural Image Translation for Human Figures	[Report]
› **AI: Ethics, Policy & Society**	Implicit Association Test in Multimodal Generative Models	[Report]

Teaching Assistantship

› **CS 484 Theory/Application Data Mining, GMU**	*Instr.: Keren Zhou*	Spring '26
› **CS 584 Theory/Application Data Mining, GMU**	*Instr.: Ziwei Zhu*	Fall '25
› **CS 108 Introduction to Computer Programming, GMU**	*Instr.: Sara Lynn Gopalakrishna, Bella Yin*	Spring '24
› **CS 478 Natural Language Processing, GMU**	*Instr.: Antonios Anastasopoulos*	Fall '23
› **COMP 502 Mathematical Foundations of Computing, GMU**	*Instr.: William D Ellis*	Fall '23
› **CS 112 Introduction to Python Programming, GMU**	*Instr.: Shahnaz Kamberi*	Fall '22, Spring '23

Student Mentorship

› **Daksha Rajagopal [TJHSST → Purdue CS]**	Modeling normative ethics and cross-cultural moral reasoning
› **Angela Liu [TJHSST → Bryn Mawr College]**	Assessing moral attributional biases in ambiguous contexts in LLMs
› **Mahika Banerjee [TJHSST → KU Leuven]**	Measuring attribution bias, coauthor on the ACL Findings '26 paper
› **Mamnuya Rinki [MS Thesis, GMU → IntraFi]**	Mitigating South Asian Biases, coauthor on TrustNLP @ ACL '26 paper
› **Yongxu Sun [MS, UW → PhD, UW]**	Inoculation Prompting by in-context learning to debias LLM decisions
› **Diwita Banerjee [MS, GMU → DoorDash]**	Assessing cuisine biases; scientific figure hallucination detection
› **Hasini Jasthi [TJHSST]**	Accent and dialect disparities in voice assistants for Global South
› **Sina Mansouri [MS, GMU]**	Long Context Memory Information Retrieval in LLM Agents
› **Sharanya Garika [CS 678 Advanced NLP]**	Identifying intersectional bias in LLMs across Indo-Aryan languages
› **Aksh Patel [CS 678 Advanced NLP]**	Identifying intersectional bias in LLMs across Indo-Aryan languages
› **Samruddhi Deshmukh [CS 795 AI Ethics]**	Auditing Text-to-Image Generation Models for Social Representation
› **Abhi Krishna [CS 795 AI Ethics]**	Auditing Text-to-Image Generation Models for Social Representation

Awards & Grants

- **SAC Highlight Award at ACL, '26** — Vignette selected as top 1.5% of accepted papers.
- **Outstanding Academic Award at GMU, '26** — Received an Outstanding Academic Achievement Award.
- **Graduate Teaching Award at GMU, '26** — Received an Outstanding Graduate Teaching Assistant Award.
- **Outstanding Reviewer Award at EACL '26** — Received an Outstanding Reviewer Award at EACL '26.
- **CAHMP Fellowship at GMU, '24** — Summer research on social biases in question answering.
- **Best Paper Award at MASC-SLL '24** — A Psychological View to Social Bias in LLMs.
- **Summer Graduate Research Award at GMU, '23** — Explainability and Robustness of Debiasing.
- **OpenAI Researcher Access Program Award '25** — Funded USD 1,000 worth of OpenAI API credits.
- **Travel Grant by AAAI/ACM AIES '24 & '23** — Funded USD 2,000 to attend AIES in San Jose and Montreal.
- **Research Excellence Award at DTU '23 & '22** — Received three Commendable Research Awards with INR 150,000.
- **Graduate Scholarship at DTU '19** — Received INR 297,000 by AICTE for qualifying GATE.

Talks

Rethinking Social Bias in NLP	*Invited Talk, MilaNLP, Fall '26*	Bocconi, Remote
Ethics and Bias in NLP	*Invited Lecture, CS 678 Advanced NLP, Fall '26*	George Mason, VA
VIGNETTE	*Talk at ACL '26*	San Diego, CA
LLM Ethics: Bias and Mitigation	*Invited Lecture, CS 678 Advanced NLP, Fall '24*	George Mason, VA
LLM Ethics: Bias and Mitigation	*Invited Lecture, CS 782 Advanced ML, Spring '24*	George Mason, VA
Breaking Bias, Building Bridges	*Talk at AAAI/ACM AIES '24*	San Jose, CA
Breaking Bias, Building Bridges	*Talk at SouthNLP '24*	Emory, GA
Breaking Bias, Building Bridges	*Talk at MASC-SLL '24*	Johns Hopkins, MD
Research Seminar	*Invited Talk, CS Department, Summer '24*	George Mason, VA
Ethics in NLP	*Invited Lecture, CS 478 NLP, Fall '23*	George Mason, VA
Text Tokenization and Probability	*Invited Lecture, CS 478 NLP, Fall '23*	George Mason, VA
True and Fair	*Talk at AAAI/ACM AIES '23*	Montreal, QC
Cyberbully Monitoring	*ML4AU Research Showcase '21*	Remote

Program Committee

- **Reviewer** — ARR '23-'26, NeurIPS '26, CHI '26, AAAI '26, COLM '25-'26, TrustNLP '24-'26, ACL Demo '26, NAACl Demo '26, EACL Demo '26, IASEAI '26, LTEDI @ EACL '24, AIES '23-'26, CSCW '24, NeurIPS '23, ACM TIST

References

- **Dr. Ziwei Zhu** — *Assistant Professor, George Mason University*
- **Dr. Antonios Anastasopoulos** — *Associate Professor, George Mason University*
- **Dr. Aylin Caliskan** — *Associate Professor, University of Washington*